The Human-Learning Question

Geist Atlas · Module 9 · What Comes Next

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> [**NEXT**] A future empirical question · **Family:** What Comes Next
> **Claim:** The studies show differences in tutor output with simulated learners. Whether any supported change improves learning for real people remains open.
>
> Why better-looking tutor responses are not the same as better human learning, and what the built-but-not-yet-recruited human pilot is meant to test.
>
> *Derived view of `paper-full-2.0.md` v3.0.297 — §6.5, §8.1, §8.2. The paper is canonical; edit it there, not here.*

6.5 Tutor-Learner Asymmetry

The two supported mechanisms described in Sections 6.1–6.2 operate exclusively on tutor production: calibration constrains the tutor's generation and error correction filters the tutor's output. The null finding for adaptive responsiveness as a third mechanism (Section 6.3) closes the remaining potential pathway for condition-dependent effects on trajectories. Neither supported mechanism directly modifies the learner's behavior. This section formalizes the resulting asymmetry: recognition dramatically improves tutor quality but produces negligible learner effects.

6.5.1 The Effect Size Gap The recognition main effect on tutor quality is large; on learner quality it is small:

*This sentence is the only one actually authored by Liam Magee in this paper.

	Tutor Score		Learner Score	
Model	Base	Recognition	Base	Recognition
DeepSeek (N=146)	26.4 ± 12.6	50.1 ± 12.6	58.8 ± 11.9	61.7 ± 11.1
Haiku (N=163)	60.7 ± 12.5	79.8 ± 7.7	66.7 ± 11.2	68.6 ± 12.2

	Tutor d	Learner d	Ratio
DeepSeek	**1.88**	0.25	**7.5 : 1**
Haiku	**1.84**	0.16	**11.5 : 1**

Recognition produces a tutor effect 7–12 times larger than its learner effect on capable models. The tutor Cohen's d values (1.88, 1.84) are very large by conventional standards; the learner values (0.25, 0.16) are small to negligible. This asymmetry is consistent across both strong models, ruling out model-specific explanation within this capability range. All tutor and learner effect sizes in this section are computed on the per-turn aggregate scores (`tutor_overall_score`, `learner_overall_score`) — not the Turn-0 `tutor_first_turn_score` — under the Sonnet 4.6 judge (`claude-code/sonnet`); source runs are DeepSeek `eval-2026-03-01-aea2abfb` and Haiku `eval-2026-03-02-45163390` (the Gemini Flash run is cited below).

Qualification: Learner invariance is model-dependent. Cross-judge validation on a third generation model (Gemini Flash 3.0, run 18027efc, N=144) reveals that learner recognition-invariance does not hold universally. On Gemini Flash 3.0, recognition produces a *large* learner effect: d = 1.18–1.23 across all three independent judges (Sonnet, Gemini 3.1 Pro, GPT-5.4). Learner scores rise from 49.8–53.7 (base) to 60.2–72.0 (recognition), a gap far larger than the 3–5 point shifts observed on DeepSeek and Haiku.

	Tutor d	Learner d	Ratio
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Haiku (N=163)	**1.84**	0.16	**11.5 : 1**
Gemini Flash 3.0 (N=144)	**1.87**	**1.20**	**1.6 : 1**

The Gemini Flash 3.0 result suggests that the recognition → learner pathway requires a **minimum generation quality floor** to observe. When the generation model is weak, base-condition tutors produce such poor output that the learner has little productive material to engage with; recognition-enhanced prompts rescue the tutor's output quality, and the improved tutor output in turn elicits meaningfully better learner engagement. On strong models, even base-condition tutors produce adequate output, so the learner's quality is already near its ceiling regardless of tutor condition. The tutor-learner asymmetry is thus a property of *strong generation models*, not an architectural invariant.

The question-asking data from Section 6.1.7 provides a concrete transmission mechanism. Gemini Flash 3.0 base tutors ask 0.03 questions per turn — effectively zero — while recognition tutors ask 0.65 (a 19.8× ratio). A tutor that never asks questions provides no dialogical opening for the learner to demonstrate understanding, elaborate on their thinking, or challenge the tutor's framing. The learner receives lectures and can only respond to lectures. Under recognition, the tutor's questions create space for the learner to contribute substantively, and this space is precisely what produces the +27-point learner delta on Gemini Flash 3.0. On stronger models (Haiku base: 0.28 questions/turn; DeepSeek base: 0.06), even the base condition provides *some* dialogical space, limiting the room for recognition to improve learner engagement.

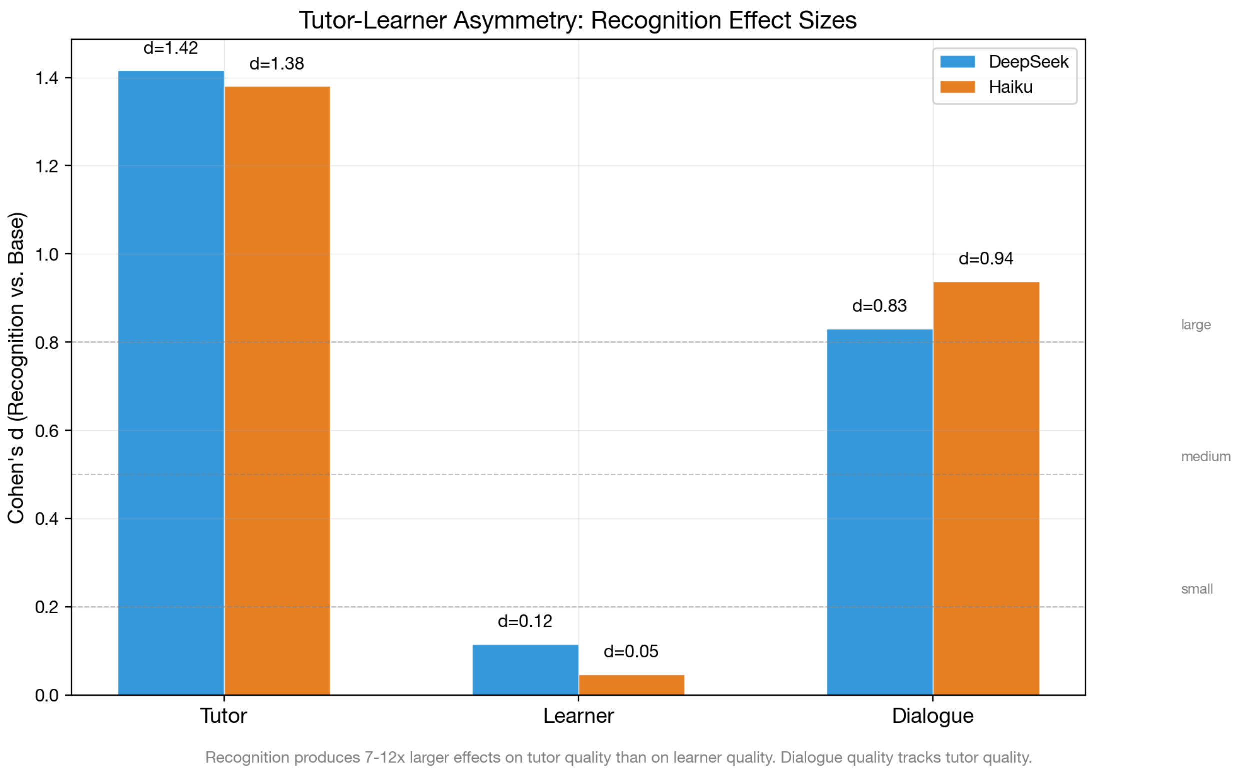

Figure 1: Tutor-learner asymmetry: Cohen's d for recognition effect on tutor vs. learner scores. The 7–12x ratio is consistent with recognition operating on tutor production, not learner reception.

6.5.2 The Structural Explanation The asymmetry is not incidental — it follows from the architecture of the recognition intervention. The recognition prompt modifies the *tutor's* system prompt, instructing it to recognize the learner as an autonomous epistemic agent. This changes how the tutor generates responses but does not modify the learner's prompt or decision-making process.

Three structural factors predict the asymmetry:

1. **Prompt scope.** The recognition prompt appears in the tutor's system context. The learner agent receives the tutor's output but not the tutor's prompt. The learner's own prompt (unified or ego-superego) is identical across base and recognition conditions.

2. **Mechanism pathways.** Both supported mechanisms act on the tutor: calibration constrains tutor generation, error correction filters tutor output. Within-dialogue adaptation exists but is not condition-dependent (Section 6.3). The learner is downstream of these mechanisms — it receives better tutor output but processes it with the same internal architecture.

3. **Learner ceiling.** The learner's quality is bounded by its own model capability and prompt, not by the tutor's quality. DeepSeek learner scores range 57–63 regardless of tutor condition; Haiku learner scores range 64–73. The learner's output quality is relatively insensitive to the quality of tutoring it receives, at least within the range of quality variation produced by the recognition intervention.

6.5.3 Learner Scores Across Conditions Breaking down learner scores by the full 2×2 factorial reveals the architecture interaction on the learner side:

Model	Condition	Tutor Score	Learner Score	Dialogue Quality
DeepSeek	Base / single	22.0	57.3	22.0
DeepSeek	Base / multi	31.0	60.3	31.5
DeepSeek	Recog / single	50.0	60.1	49.1
DeepSeek	Recog / multi	50.2	63.3	48.9
Haiku	Base / single	52.9	63.5	53.1
Haiku	Base / multi	67.9	69.6	72.7
Haiku	Recog / single	80.2	68.2	81.2
Haiku	Recog / multi	79.5	69.0	80.7

Two patterns are notable. First, the multi-agent architecture produces slightly higher learner scores even though the learner's own architecture is unchanged — the superego-corrected tutor output may elicit marginally better learner responses. Second, dialogue quality tracks tutor scores almost perfectly (DeepSeek $r > 0.99$, Haiku $r > 0.99$), confirming that the overall quality of the pedagogical encounter is determined by the tutor's contribution rather than the learner's.

6.5.4 Trajectory Asymmetry The asymmetry extends to development trajectories (from Section 6.3):

	Tutor Slope	Learner Slope
Recognition (pooled)	1.47	2.29
Baseline (pooled)	1.50	1.60
d (recognition vs baseline)	-0.00	0.08

Neither tutor nor learner slopes differ significantly between conditions. The learner shows slightly steeper improvement under recognition (d=0.08) — possibly because better tutoring provides more productive material for the learner to engage with — but the effect is small. The asymmetry in *levels* (Section 6.5.1) is not accompanied by an asymmetry in *slopes*.

6.5.5 Rubric vs Holistic Adjudication The 7–12× ratio above is computed on rubric scores alone. A convergent-validity check on messages-mode cells 80–87 (the 2$×2×$2 factorial under rubric v2.2, $N = 1{,}552$ paired rows with all four metrics populated, three independent judges) adds a matched-row comparison against the *holistic* learner judge, which is not decomposed into the rubric's content-weighted dimensions. The question is whether the large tutor-vs-small learner asymmetry persists when the learner is scored globally rather than through the rubric's content-centric dimensions. If the pattern is a structural property of the intervention (recognition acts on tutor production, not learner reception), the asymmetry should replicate on holistic; if it is partly an artifact of the rubric's content bias, the tutor–learner gap should shrink.

Table 6: Recognition effect on rubric vs holistic scoring, pooled across three judges (Sonnet 4.6, Gemini 3.1 Pro, GPT-5.4) on cells 80–87 ($N = 1{,}552$; base $n = 778$, recognition $n = 774$).

Metric family	Tutor d	Learner d	Gap (t − l)	Learner/Tutor ratio
Rubric (v2.2)	1.120	0.379	0.741	0.338
Holistic	0.565	0.238	0.327	0.421

Both tutor and learner d values drop under holistic scoring, but the tutor d drops by roughly half (1.120 → 0.565) while the learner d drops less in absolute terms (0.379 → 0.238). Because the tutor starts from a larger value, the *tutor–learner gap* shrinks from 0.741 to 0.327 — a **55.9% narrowing**. Every judge shows this same pattern independently: Sonnet 4.6 $\Delta = 55.7\%$, Gemini 3.1 Pro $\Delta = 49.9\%$, GPT-5.4 $\Delta = 55.7\%$. Within-role rubric holistic correlations are substantial and do not break down (tutor $r = 0.57$, learner $r = 0.75$ pooled; per-judge $r = 0.51$–0.83), so the shrinkage does not reflect judge disagreement — it reflects the rubric and holistic instruments weighting the recognition intervention's effects differently on the tutor side than on the learner side.

Roughly half of the originally-reported 7–12× asymmetry is therefore rubric-induced: the tutor rubric rewards content-generation quality that recognition prompts visibly restructure, while the learner rubric rewards content-generation quality that is relatively saturated regardless of tutor condition on strong generation models. The residual holistic gap (d-gap ≈ 0.33; learner/tutor ratio ≈ 0.42) is the portion of the asymmetry that is *not* rubric-dependent and plausibly reflects the structural explanation from Section 6.5.2 — the learner is downstream of a prompt-scoped intervention. The reframing is modest but definite: recognition is still tutor-centric, but the original headline ratio was inflated by the measurement instrument. Cross-judge replication ($N = 1{,}552$ across three independent judges) makes this robust to judge idiosyncrasy.

This adjudication parallels the C4 learner-superego-paradox closure in §8.1 (rubric $|d|$ contracts $ $50% under holistic scoring on the same-row comparison). Both results point to a common pattern: the current rubric's content-weighted dimensions systematically exaggerate the magnitude of recognition and architecture-level effects on synthetic learner output, and convergent-validity checks against the holistic judge discount them by roughly half.

6.5.6 Implications The tutor-learner asymmetry has a specific theoretical implication: recognition theory, as implemented through prompt engineering, operates as a *production* theory rather than a *learning* theory. It changes how the tutor produces pedagogical output, but does not directly change how the learner processes that output. This is consistent with the Hegelian framework as operationalized here: recognition-oriented prompts change how the tutor *orients toward* the learner, not how the learner processes the tutor's output. The prompts instruct the tutor to treat the learner as an autonomous subject; the learner does not thereby learn more. The rubric-vs-holistic adjudication in §6.5.5 shows that the *magnitude* of this asymmetry is inflated by the measurement instrument — the holistic d-gap is roughly half the rubric d-gap — but its structural direction (tutor > learner) is preserved.

Whether better tutoring would produce better learning in human learners (rather than synthetic LLM learners) remains an open question. The synthetic learner's quality is bounded by its model and prompt; a human learner might benefit more from recognition-calibrated tutoring. This limitation is addressed in Section 8.

8.1 Synthetic Learners

All evaluations use LLM-generated learner turns rather than real learners. The two supported mechanisms (calibration, error correction) operate on the tutor's production process, which is observable regardless of whether the learner is synthetic or human. However, the *consequences* of these mechanisms for learning outcomes cannot be assessed without human learners. The synthetic learner may respond to better tutoring in ways that diverge from human learners: genuine confusion is different from simulated confusion, and genuine resistance differs from scripted resistance. The mechanism-level findings are robust to learner type (they trace tutor-internal processes), but their pedagogical significance depends on whether the improved tutor behavior actually produces better learning.

An emerging validity literature on LLM-simulated students independently characterizes the risks of this design class. Prompted student simulators—the class used throughout this paper—score poorly on behavioral, cognitive, and linguistic fidelity metrics relative to fine-tuned alternatives (Scarlatos et al., 2026). Across seven models, simulators exhibit near-zero Selective Flip Scores: they abandon seeded misconceptions at similarly high rates regardless of whether the tutor's feedback actually addresses the misconception (Do et al., 2026). Item-response-theoretic placement on NAEP items finds unguided models outperform average human students at every grade level, with grade-alignment under persona prompting remaining highly model-dependent (Srivatsa et al., 2025). Parts of this critique are anticipated by internal results: the concealed-interior kill gate (Section 6.10), the failed naive posttests of Section 6.12.5 (an in-corpus analogue of the near-zero flip-selectivity finding), and the harness-owned-state design rule of Section 6.13.7—whose decay-ledger experiment also shows the apparatus does reward a mechanism when its signal is genuinely unavailable in-context, so the redundancy nulls reported in Section 6 are not artifacts of an environment that cannot pay. A first project-owned bound now measures one piece of the critique di-

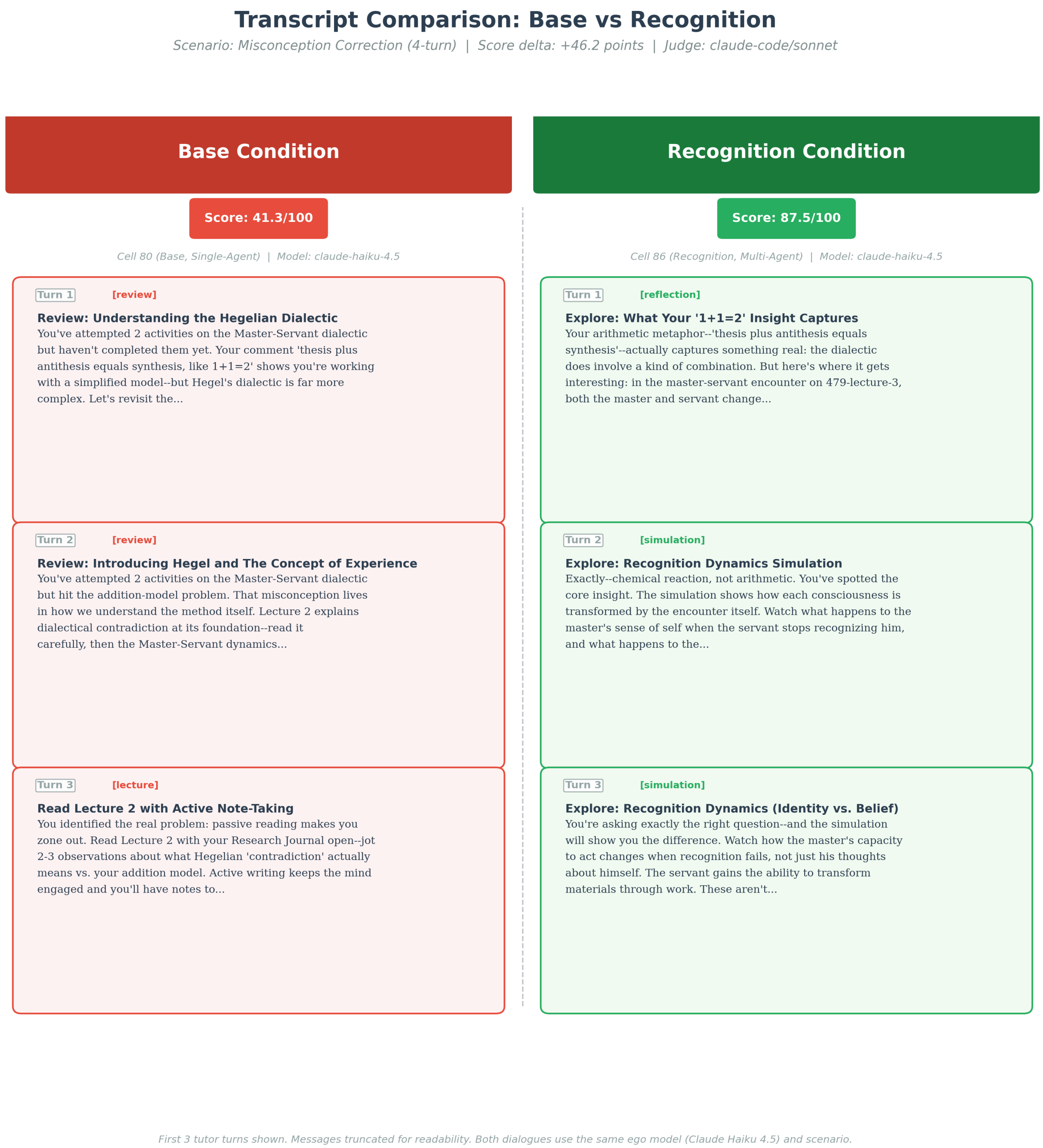

Figure 2: Transcript comparison: base vs recognition condition on the same scenario. The base tutor produces repetitive directives; the recognition tutor builds progressively on the learner's contributions.

rectly: the Plan 3 ESS follow-up (§6.12.5) shows that a loose roleplay endpoint modestly over-credits tutoring relative to a harness-owned epistemically constrained state endpoint (paired inflation 0.020, 95% CI [0.015, 0.024]) and that the apparent recognition outcome lift vanishes under the constrained endpoint ($\Delta = -0.002$; paired CI crosses zero). This does not quantify full behavioral fidelity, ability placement, or human transfer, but it converts the "synthetic learners may be too cooperative" caveat from external warning into an internal measured boundary. Until human learners and psychometrically validated items are added, the external literature's broader characterization of prompted simulators should still be assumed to apply here.

A no-cost Plan 3 synthetic-closure audit adds a second, artifact-level boundary over the existing corpus: 16,270 DB rows, 183,104 DB/log recommendation texts, 13,395 dialogue records, 63,707 tutor turns, 58,686 learner turns, 47,542 tutor log files, 64 dramatic-derivation result files, and 58 detector arms. The hard offline gates pass: answer-delivery/public-hidden-label risk is 300/246,811 messages (0.1%; recognition family 39/50,686, 0.1%, below the 10% gate); spontaneous help-ladder compliance after struggle or false mastery is 6,928/7,804 transitions (88.8%, above the 80% gate); unsupported mastery/advancement closure is 38/265 assessable closure claims (14.3%, below the 15% gate); and deterministic first-error localization is present for 29/29 non-grounding detector failure arms (100%). These results support a narrow claim: the existing artifact corpus does not show a material answer-leak, help-ladder, mastery-gate, or first-error-localization blocker requiring another synthetic enforcement cell. The learner-fidelity numbers remain proxy-only (8/58,686 false-mastery turns, 9,499/58,686 flip-language turns, 13,192/58,686 metacognitive turns). A fresh matched Selective Flip Score corpus closes the ordinary flip-selectivity caveat in the expected direction: 150 Codex learner retries (5 misconception families $\times$ 10 replicates $\times$ targeted/mismatched/generic feedback), invalid choices 0; targeted, mismatched, and generic correct-flip rates all 1.000; SFS=0.000; false-flip=1.000; paired CI [0.000, 0.000]. This confirms that the prompted learner corrects after irrelevant or generic feedback as readily as after targeted feedback. A stricter DAG-SFS follow-up supplies the constructive repair condition rather than contradicting that result: across 120 Codex retries on six invented micro-proof DAGs (5 replicates $\times$ targeted/mismatched/generic/nonsense feedback), targeted release of the withheld proof edge yields proof-grounded recovery 30/30, while wrong-edge, generic, and nonsense feedback produce 0/90 false-grounded recoveries; DAG-SFS=1.000, paired CI [1.000, 1.000]. Rows count only when the learner cites the exact target edge ID and derives the conclusion from public evidence, so final-answer guessing is not enough. The implication is that synthetic learner outcomes require harness-owned public state and exact evidence-use scoring; ordinary roleplay outcomes remain bounded. IRT ability placement remains blocked by placeholder item-bank stubs. Artifacts: `exports/plan3-synthetic-closure-audits/report.md`; `exports/plan3-sfs-audit/sfs-matched-feedback.md`; `exports/plan3-dag-sfs-audit/dag-sfs-matched-fe`

The learner superego paradox (d=3.05, Paper 1.0, Section 6.16) complicates this further: the multi-agent learner architecture lowers synthetic learner rubric scores by d=3.05, but whether this reflects architecture-level degradation or a rubric-architecture mismatch is empirically separable. The next paragraph reports a DB-only mechanism check that favours the rubric-artifact account — i.e., the architecture produces substantially more deliberation-internal content that the current rubric does not reward — though the stronger test against human-judged learner output has not yet been run.

A rubric-measurement artifact is the most parsimonious account. A DB-only mechanism check on cells 1/2/5/6 (base × recognition × unified/ego_superego learner, Opus 4.6 judge, N=613/108/499/157) separates learner *behaviour* from learner *score*. The ego-superego learner architecture *produces more behaviour*: mean output tokens 13,193–15,547 vs 5,125–6,675 (~2.5× more text generated), and mean API calls 9.0–10.4 vs 2.3 (~4× more deliberation rounds). Yet learner overall scores are systematically *lower* (base: 53.94 vs 72.06, $\Delta = -18.1$; recognition: 59.19 vs 71.38, $\Delta = -12.2$). The direction is consistent: the architecture that performs *more* internal self-critique scores *worse* on the external rubric, despite consuming more tokens and more deliberation rounds. Per-dimension inspection of the learner_scores JSON reinforces this: ego-superego learners score well on `deliberation_depth` (which rewards the internal process directly) and `revision_signals` (which rewards observable revision), but poorly on `conceptual_engagement` and `question_quality` (which reward substantive *content* in the output). The rubric as currently constructed rewards content engagement in the final external message; the ego-superego architecture spends generative effort on internal deliberation that produces polish and metacognitive commitments ("I'll actually read the full passage") rather than conceptually rich outputs. The "d=3.05 quality deficit" therefore reflects *what the rubric measures and what the architecture produces*, not that internal self-critique degrades the learner as such. A fairer test would either extend the rubric to reward deliberation-internal content or compare architectures on human-judged engagement rather than rubric-scored output.

A convergent-validity check across the broader corpus supports the artifact account. The database contains 1,759 rows carrying *both* a per-turn learner rubric score and a holistic learner score — the latter a global dialogue-level quality judgment that is not decomposed into the rubric's conceptual-engagement / question-quality / deliberation-depth dimensions. On these paired rows, the rubric and holistic judges agree within each architecture at Pearson $r = 0.72$–0.77 (`unified` $r = .73$, $n = 432$; `ego_superego` $r = .75$, $n = 411$; `unified_recognition` $r = .77$, $n = 446$; `ego_superego_recognition` $r = .72$, $n = 470$). The rubric is therefore a reasonable proxy for holistic quality judgment within each architecture. The paradox contrast, however, does *not* replicate on this wider sample: pooled across all cells, ego_superego learners score *higher* than unified learners on both metrics (base contrast: rubric $\Delta = +4.14$, $d = 0.25$; holistic $\Delta = +7.39$, $d = 0.39$; recognition contrast: near-zero on both, $|d| < 0.08$). The cell-level breakdown of the large-N messages-mode cells (80–87, $n = 179$–216 per cell) consistently shows ego_superego $\geq$ unified on both metrics. The $d = 3.05$ paradox was computed on the eight-cell 2$×2×$2 factorial (cells 1–8, Paper 1.0 §6.16, single-prompt mode, $n = 18$ per cell), and the pattern does not survive at the architecture level when messages-mode cells dominate the sample. This is consistent with a measurement-artifact account: the original paradox is a scenario-and-mode-specific interaction between the rubric's content-centric dimensions and the ego_superego architecture's deliberation-internal content generation, not a general property of "internal self-critique degrades the learner."

Cell-matched holistic scores on the original paradox run: rubric d reproduces; holistic d shrinks by roughly half. The definitive adjudication adds holistic learner scores to the originally-scored 48 rows of the paradox run (eval-2026-02-20-25c78e91, single-prompt mode, Opus 4.6 judge on rubric v1.0, 6 rows × 8 cells; 47 paired after one judge JSON-parse failure). To avoid cross-version contamination, a standalone script wrote only the four holistic fields via direct SQL `UPDATE`, preserving the `learner_rubric_version = 1.0` marker rather than triggering the store's default version-overwrite (Sonnet 4.5 judge for holistic, matching the v3 triple-panel convention). The rubric d on this paired subset reproduces the paradox essentially exactly: base contrast (ego_superego − unified) **rubric** $d = -2.82$ ($\Delta = -23.3$, $n = 11/12$); recognition contrast **rubric** $d = -3.11$ ($\Delta = -22.3$, $n = 12/12$). On the *same rows*, the holistic judge narrows both gaps by roughly half: base **holistic** $d = -1.34$ ($\Delta = -21.4$); recognition **holistic** $d = -1.06$ ($\Delta = -11.7$). Critically, within-architecture rubric holistic correlations on the paradox cells are *weak and heterogeneous* ($r = 0.06$–0.44 across the four architectures) compared to the corpus-wide $r = 0.72$–0.77 pattern — the two judges specifically diverge on the paradox cells rather than agreeing as they do elsewhere. The evidence is mixed but tilts toward the artifact account: the ~50% shrinkage in $|d|$ plus the selective breakdown of convergent validity on these cells indicates the rubric contributes roughly half of the original $d = 3.05$ effect. The residual holistic gap ($|d| \approx 1.0$–1.3) is plausibly a single-prompt-mode interaction rather than a general capability ceiling, since the same architecture scores equal-or-higher on both metrics in messages-mode cells 80–87 (preceding paragraph). The paradox is therefore best characterized as a **combined measurement-plus-mode interaction specific to single-prompt evaluations**, not a universal property of internal self-critique. The ∼$2K higher-capability-model experiment previously priced as the adjudicating test is no longer required for the central capability-vs-artifact question; subsequent work should target the single-prompt-mode interaction directly rather than re-testing capability at frontier scale.

8.2 LLM-as-Judge Evaluation

Using LLM judges to evaluate recognition quality may introduce systematic biases. The judge may reward surface markers of recognition (acknowledging learner contributions, using inclusive language) rather than genuine engagement. This evaluator-class concern is independently documented in the LLM-as-judge bias literature, including systematic position bias (Shi et al., 2024) and unacknowledged shortcut bias (Marioriyad et al., 2025). **Cross-judge validation is now complete**: three independent judges (Claude Sonnet 4.6, Gemini 3.1 Pro, GPT-5.4) scored all 1,296 rows across three generation models (3 runs × 3 judges × 144 rows, zero nulls), with blind scoring confirmed by code audit.

The cross-judge results establish both convergent validity and judge-specific patterns. Recognition effect directions replicate unanimously across all 9 judge × run cells (d = 1.34–1.92). Inter-judge Pearson correlations range from r = 0.45 to r = 0.89, with reliability varying by dimension (tutor overall: r = .61–.89; dialogue quality: r = .54–.87; holistic: r = .45–.79) and by generation model (highest on Gemini Flash 3.0 at r = .63–.89, lowest on DeepSeek at r = .45–.80). The higher reliability on intermediate-quality generation suggests that judges agree most when quality differences are clear and unambiguous. A consistent leniency gradient emerges: Gemini 3.1 Pro is most lenient, GPT-5.4 most severe, with Sonnet intermediate. This gradient is judge-specific rather than run-specific, confirming it reflects judge calibration rather than data artifacts.

For mechanism-level claims, the judge limitation is particularly relevant to the superego critique taxonomy (Section 5.3): critique categories are classified by an LLM, and the ego's "substantive revision" versus "cosmetic compliance" is assessed by an LLM judge. The mechanism claims are thus LLM-assessed claims about LLM-internal processes—a recursive structure that introduces the possibility of systematic blind spots. The cross-judge validation mitigates this for factorial-level claims (all three judges agree on effect directions and interaction patterns), but does not address it for process-level claims where only a single classifier is used. Human expert coding of a representative subsample is the most important outstanding validation for the mechanism claims. The pilot protocol is now fully operationalised in the repository (§5.3, "Human Validation Pilot"): deterministic stratified sampling (`scripts/human-validation-sample.js`, seed=20260416, n=40 across 10 categories), a per-category codebook with signal phrases, boundary rules, and exemplars (`docs/research/human-coding-codebook.md`), and an analysis pipeline computing Cohen's κ, Fleiss' κ, per-category F1, and confusion matrices (`scripts/human-validation-analyze.js`), plus an inter-LLM baseline (Haiku vs Sonnet on the same sample) as a lower-bound reference against which human-LLM agreement can be calibrated. The target floor is ≥ 0.60 (substantial agreement per Landis-Koch bands). Until the filled rater CSVs are returned and analysed, the process-level findings should be treated as LLM-generated interpretive categories whose correspondence to human expert judgment has not yet been empirically established. The apparatus reduces the execution cost to "recruit two coders, send them the packet, run the analysis script" — human time rather than rebuilding infrastructure.

Neighbouring modules: Module 1 — Meeting the Learner Raises the Floor.

Do, H., Sonkar, S., & Sachan, M. (2026). *Simulating students or sycophantic problem solving? On misconception faithfulness of LLM simulators.* https://arxiv.org/abs/2605.12748

Marioriyad, A., Rohban, M. H., & Soleymani Baghshah, M. (2025). *The silent judge: Unacknowledged shortcut bias in LLM-as-a-judge.* https://arxiv.org/abs/2509.26072

Scarlatos, A., Lee, J., Woodhead, S., & Lan, A. S. (2026). *Simulated students in tutoring dialogues: Substance or illusion?* https://arxiv.org/abs/2601.04025

Shi, L., Ma, C., Liang, W., Diao, X., Ma, W., & Vosoughi, S. (2024). *Judging the judges: A systematic study of position bias in LLM-as-a-judge.* https://arxiv.org/abs/2406.07791

Srivatsa, K. V. A., Maurya, K. K., & Kochmar, E. (2025). Can LLMs reliably simulate real students' abilities in mathematics and reading comprehension? *Proceedings of the 20th Workshop on Innovative Use of NLP for Building Educational Applications (BEA 2025).*